User Acceptance Testing (UAT) Template

Date	12-10-2026
Team ID	SWUID2026403484
Project Name	HotelPro - Hotel Room Booking and Management System
Maximum Marks	

Project Overview:

Project Name: HotelPro - Hotel Room Booking and Management System

Description: A web-based MERN-stack (MongoDB, Express.js, React.js, Node.js) application that lets customers search hotels and rooms, check availability, book and pay online, while hotel staff and administrators manage hotels, rooms, guests, bookings, check-in/check-out and payments. It uses JWT authentication with role-based access control for customers, staff and administrators, and provides an administrator dashboard with booking, occupancy and revenue statistics.

Project Version: v1.0.0

Testing Period:2027-01-11 to 2027-01-17 (planned, after Sprint 6)

Testing Scope:

- User registration, login, logout, profile and password reset
- JWT authentication and role-based access (customer, staff, admin)
- Hotel and room management by administrators
- Room search, filters and availability checking
- Room booking, price calculation and double-booking prevention
- Online payment, offline (cash) payment, receipts and refunds
- Booking management: view, modify, cancel and status tracking
- Guest management, check-in and check-out by staff
- Admin dashboard, user management and reports
- Security, responsive UI and browser compatibility

Requirements to be Tested:

- As a customer, I want to search for rooms by city, dates and guests and see only available rooms (FR-SRC).
- As a customer, I want to book a room and pay securely without the room being double-booked (FR-BKG, FR-PAY).
- As a customer, I want to view, modify or cancel my bookings and receive a refund where eligible (FR-BMG, FR-PAY-05).
- As hotel staff, I want to manage guest records and perform check-in and check-out (FR-GST, FR-CIO). • As an administrator, I want to manage hotels, rooms and users and view statistics and reports (FR-HRM, FR-ADM).
- As any user, I want secure login with access limited to my role (FR-REG, FR-JWT).

Testing Environment:

- **URL:** http://localhost:3000 (replace with the deployed or localhost URL)

- **Credentials (demo accounts):**
- **Customer:** test.customer@example.com / Test@1234
- **Hotel Staff:** test.staff@example.com / Test@1234
- **Admin:** admin@example.com / Admin@1234
- **Payment test inputs:** use the sandbox test card/UPI values of the integrated payment gateway (one that succeeds and one that fails). Fill in once the gateway is finalised.
- **Test data needed:** at least 2 hotels in different cities; rooms of each type (single, double, deluxe, suite); one room under maintenance; one room with an upcoming confirmed booking; one booking in each status (pending, confirmed, checked-in, checked-out, cancelled).
- **Browsers/devices:** latest Chrome, Firefox, Edge and Safari; one mobile and one tablet viewport.

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
A. Registration, Login and Authentication					
TC-001	User registration (valid) *FR-REG-01, 02*	1. Open site and click Register 2. Enter name, unique email, phone and a password of 8+ characters 3. Submit	Account created with role customer; user can log in; no password shown in any response		
TC-002	Registration with existing email (negative) *FR-REG-02*	1. Open Register 2. Enter an already registered email with other valid details 3. Submit	Error shown that email already exists; no duplicate account created		
TC-003	Invalid email format / weak password (negative) *FR-REG-02*	1. Open Register 2. Enter email without @ and a 5-character password 3. Submit	Inline validation errors for email format and minimum password strength; form not submitted		
TC-004	Role cannot be chosen at registration (negative) *FR-REG-01, FR-JWT-05*	Register through the UI, then resend the register API call in Postman with role = admin	Account is created as customer only; role value from client is ignored or rejected		
TC-005	Password stored as bcrypt hash *FR-REG-03*	1. Register a user 2. Open the users collection in MongoDB	password field holds a bcrypt hash (starts with $2a/$2b), not plain text; password never returned by any API		
TC-006	Login (valid) *FR-REG-04, FR-JWT-01*	1. Open Login 2. Enter registered email and correct password 3. Click Login	User logged in and redirected to home; name shown in header; JWT issued containing user ID and role		
TC-007	Login with wrong password / unknown email (negative) *FR-REG-05*	1. Login with valid email and wrong password 2. Login with an unregistered email	Same generic error (for example "Invalid email or password") in both cases; does not reveal which field is wrong		
TC-008	Logout *FR-REG-06*	1. Login 2. Click Logout 3. Press browser Back / open My Bookings URL	Session ends on client, token removed; protected pages redirect to Login		
TC-009	View and update profile *FR-REG-07*	1. Login 2. Open Profile 3. Change name and phone and save 4. Reload page	Changes saved and shown after reload; email and role cannot be altered from profile		

TC-010	Password reset via email *FR-REG-08*	1. Click Forgot password 2. Enter registered email 3. Open the reset link/code from the email and set a new password 4. Login with old, then new password	Reset succeeds; old password rejected; new password works		
TC-011	Rate limiting on login (negative) *NFR 4.2*	Submit wrong password repeatedly (about 10 times quickly) for one account/IP	Further attempts blocked with HTTP 429 or lockout message for a period		
B. JWT Authentication and Role-Based Access					
TC-012	Protected API without token (negative) *FR-JWT-03, 04*	In Postman call GET /bookings/my with no Authorization header	HTTP 401 returned; no data exposed		
TC-013	Expired or tampered token (negative) *FR-JWT-02, 04*	1. Call a protected API with an expired token 2. Edit one character of a valid token and call again	HTTP 401 for both; UI redirects to Login on expiry		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-014	Token sent as Bearer header *FR-JWT-03*	1. Login and open browser developer tools 2. Perform a protected action (view My Bookings)	Request carries Authorization: Bearer <token>; response succeeds		
TC-015	Customer accessing admin features (negative) *FR-JWT-05, FR-ADM-06*	1. Login as customer 2. Open the admin dashboard URL and call an admin API (for example add hotel)	HTTP 403 / access denied page; no data exposed or changed		
TC-016	Staff accessing admin-only features (negative) *FR-HRM-06, FR-ADM-06*	1. Login as staff 2. Try to add a hotel and open the admin dashboard	Access denied (403); staff sees only staff console functions		
TC-017 Session renewal with refresh token *FR-JWT-06 (should)*		1. Login and keep session active past the access-token expiry 2. Perform an action	Session renews without forcing re-login (if refresh tokens are implemented); otherwise redirect to Login		
C. Hotel and Room Management (Admin)					
TC-018	Admin adds a hotel *FR-HRM-01*	1. Login as admin 2. Open Hotels > Add Hotel 3. Enter name, address, city, description, amenities and images 4. Save	Hotel saved and listed; visible in customer search results		
TC-019	Add hotel with missing required fields (negative) *FR-HRM-01*	1. Leave name or city empty 2. Save	Validation errors shown; hotel not created		

TC-020	Admin edits hotel details *FR-HRM-02*	1. Open an existing hotel 2. Change description and amenities 3. Save	Updated details shown on admin and customer pages		
TC-021	Admin deletes a hotel *FR-HRM-02*	1. Delete a hotel that has no bookings 2. Confirm	Hotel removed from admin list and search. (Behaviour for hotels with bookings is not defined in the SRS; record observation.)		
TC-022	Admin adds a room *FR-HRM-03*	1. Open a hotel > Add Room 2. Enter room number, type, price per night, capacity, amenities and status 3. Save	Room saved under the hotel with the entered values; appears in search for valid dates		
TC-023	Duplicate room number in same hotel (negative) *FR-HRM-03 / DB 8.2*	Add a room using a room number that already exists in the same hotel	Error shown; duplicate room not created		
TC-024	Room price and capacity validation (negative) *FR-HRM-03 / DB 8.2*	1. Add a room with price 0 or negative and capacity empty 2. Save	Validation errors (price must be greater than 0, capacity required); room not created		
TC-025	Update room status *FR-HRM-04*	1. Edit a room and set status to maintenance, then inactive, then available 2. Search for a date range	Status saved each time; rooms in maintenance/inactive status are not offered for booking		
TC-026	Delete room with active/upcoming booking (negative) *FR-HRM-05*	1. Pick a room with a future confirmed booking 2. Try to delete it	Deletion blocked with clear message; room and booking remain		
TC-027	Delete room without bookings *FR-HRM-04*	1. Delete a room that has no bookings 2. Confirm	Room removed and no longer searchable		
TC-028	Hotel/room management by non-admin (negative) *FR-HRM-06*	1. Login as customer and as staff 2. Call POST /hotels and POST /rooms APIs	HTTP 403 for both; no hotel or room created		
D. Room Search and Availability					

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-029	Search rooms (valid) *FR-SRC-01, 04*	1. Open home page 2. Enter city, check-in (tomorrow), check-out (+2 days) and number of guests 3. Click Search	List of available rooms shown with images, price per night, capacity and amenities		
TC-030	Past check-in date (negative) *FR-SRC-05*	1. Enter a check-in date in the past 2. Search	Search rejected with validation message; no results shown		
TC-031	Check-out not after check-in (negative) *FR-SRC-05*	1. Enter check-out equal to or before check-in 2. Search	Search rejected with validation message		

TC-032 Filter and sort results

1. On results page filter by price range, room type and amenities
2. Sort by price low to high and high to low

Only matching rooms displayed; ordering correct. (Rating filter depends on rating data being available; note result.)

FR-SRC-02

TC-033 Booked rooms excluded for overlapping dates

1. Book room R1 for days 10-12
2. Search days 11-13 and days 12-14 and days 8-10

R1 hidden for the 11-13 overlap; behaviour for 12-14 and 8-10 follows the system's same-day turnover rule (note result)

FR-SRC-03

TC-034 Search with no matching rooms

Search a city with no hotels, or guests above any room capacity

Friendly "no rooms found" message; no error

FR-SRC-01

TC-035 Room details page

Open a room from the results

Photos, amenities, price per night, capacity and Book Now action are displayed

FR-SRC-04

TC-036 Search response time

Run 10 searches and note response time in developer tools

Results returned in under 2 seconds under normal load

FR-SRC-06

E. Room Booking

TC-037	Book an available room *FR-BKG-01, 05, 06*	1. Login as customer 2. Select an available room and dates 3. Enter guest details and confirm	Booking created with unique booking ID and status Pending; booking summary shown; proceed to payment		
TC-038	Total amount calculation *FR-BKG-02*	1. Book a room priced 2,000/night for 3 nights 2. Check the summary	Total equals nights x price plus applicable taxes (as configured); amount matches the payment amount		
TC-039	Guest details recorded with booking *FR-BKG-04*	1. Enter guest name, contact, number of adults and children 2. Confirm booking 3. Open the booking details	All guest details saved and displayed in booking record		
TC-040	Booking without login (negative) *FR-BKG-01*	1. Log out 2. Select a room and click Book Now	User redirected to Login; booking only possible after authentication		
TC-041	Guests exceed room capacity (negative) *FR-BKG-04 / DB 8.2*	1. Select a room with capacity 2 2. Enter 3 guests 3. Confirm	Validation blocks the booking with a capacity message		

TC-042	Double booking by two users (negative) *FR-BKG-03, NFR 4.3*	1. User A and User B open the same room and dates 2. Both click Confirm at the same time (or User B confirms after A)	Only one booking is created; the other user gets a room-not-available message; no overlapping bookings in database		
TC-043	Booking confirmation email *FR-BKG-07 (should)*	1. Complete a booking 2. Check the registered email inbox	Email received with booking ID, hotel, room, dates and amount		
F. Payment Processing					

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-044	Successful online payment *FR-PAY-01, 03, 04*	1. Create a booking 2. Pay using a successful sandbox test payment method 3. Open My Bookings	Payment status success; booking status changes to Confirmed; payment record holds amount, method, transaction ID and timestamp		
TC-045	Failed payment (negative) *FR-PAY-04, NFR 4.3*	1. Create a booking 2. Pay using a sandbox test method that fails (or cancel at gateway)	Failure message shown; payment status failed; booking is NOT confirmed; user can retry payment		
TC-046	No card data stored *FR-PAY-02*	1. Complete a payment 2. Inspect payments collection and API responses	No card number, CVV or expiry stored or returned; only gateway transaction reference saved		
TC-047 Payment receipt / invoice *FR-PAY-06*		1. Complete a payment 2. Open or download the receipt	Receipt shows booking ID, amount, method, transaction ID and date		
TC-048 Staff records cash payment *FR-PAY-07*		1. Login as staff 2. Open a booking with pending payment 3. Record offline payment with amount	Payment saved with method cash; booking payment status updates; entry visible to admin		
TC-049 Duplicate transaction ID (negative) *DB 8.4*		Attempt to save two payments with the same transaction ID (via API)	Second request rejected (unique transactionId); no duplicate payment record		
G. Booking Management					
TC-050	View my bookings *FR-BMG-01*	1. Login as customer 2. Open My Bookings	Current, upcoming and past bookings listed with status, dates and amount		

TC-051	Customer cannot view others' bookings (negative) *FR-BMG-01, NFR 4.2*	1. Login as customer A 2. Call GET /bookings/{id} using customer B's booking ID	Request denied (403/404); no data of customer B returned		
TC-052	Cancel booking before check-in *FR-BMG-02, 06*	1. Open a confirmed upcoming booking 2. Click Cancel and confirm 3. Search the same room and dates	Booking status Cancelled; room appears available again for the freed dates; cancellation policy message shown		
TC-053	Cancel after check-in (negative) *FR-BMG-02*	1. Open a booking in Checked-in status 2. Try to cancel	Cancellation not allowed; message shown		
TC-054	Refund on eligible cancellation *FR-PAY-05*	1. Cancel a paid booking that is eligible under the cancellation policy 2. Check payment record	Full or partial refund recorded; payment status Refunded; amount matches policy. (Policy values are not defined in the SRS; confirm expected amount with the team.)		
TC-055	Modify booking dates (available) *FR-BMG-03*	1. Open an upcoming booking 2. Change dates to a free period 3. Save	Booking updated with new dates; total recalculated; old dates freed		
TC-056	Modify booking to unavailable dates (negative) *FR-BMG-03*	Change dates to a period when the room is already booked	Change rejected with availability message; original booking unchanged		
TC-057	Staff/Admin view, search and filter all bookings *FR-BMG-04*	1. Login as staff 2. Open Bookings 3. Filter by status, date and guest name	Correct bookings listed for each filter		
TC-058	Booking status lifecycle *FR-BMG-05*	1. Create booking (Pending) 2. Pay (Confirmed) 3. Check-in (Checked-in) 4. Check-out (Checked-out)	Status moves through Pending, Confirmed, Checked-in, Checked-out in order; each change visible to staff and customer		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
H. Guest Management (Staff)					
TC-059	Add guest record *FR-GST-01*	1. Login as staff 2. Open Guests > Add Guest 3. Enter name, contact, address, ID proof type and number 4. Save	Guest record saved and shown in guest list		
TC-060	Search and edit guest *FR-GST-01*	1. Search guest by name 2. Open record, edit contact and save	Guest found; updated details saved		
TC-061	Link guest to booking and add companions *FR-GST-02, 03*	1. Open a booking 2. Attach the guest record 3. Add an additional accompanying guest	Guest linked to booking; companion shown with the primary guest		

TC-062 Guest data access restricted (negative) *FR-GST-04*		1. Login as customer 2. Call guest list API / open staff guest page	Access denied (403); guest personal data not exposed		
TC-063 Guest stay history *FR-GST-05 (low)*		Open a guest who has stayed more than once	Past stays listed with dates and hotel		
I. Check-in and Check-out (Staff)					
TC-064 Check-in against confirmed booking *FR-CIO-01, 02, 05*		1. Login as staff 2. Open a confirmed booking for today 3. Click Check-in	Booking status Checked-in; room status Occupied; actual check-in date and time recorded		
TC-065 Check-in for unpaid/pending or cancelled booking (negative) *FR-CIO-01*		Try to check in a Pending booking and a Cancelled booking	Check-in blocked with message; statuses unchanged		
TC-066	Check-out with payment complete *FR-CIO-03, 04, 05*	1. Open a checked-in booking with full payment 2. Click Check-out	Booking status Checked-out; room set back to Available (or cleaning/maintenance if used); actual check-out time recorded		
TC-067	Check-out with outstanding balance *FR-CIO-04*	1. Open a checked-in booking with part payment 2. Click Check-out	System shows outstanding amount and requires payment or records the balance before completing		
TC-068	Late check-out / early check-in charges *FR-CIO-06 (low)*	Check out after the standard check-out time	Extra charge calculated and displayed (if implemented)		
J. Admin Dashboard and User Management					
TC-069	Dashboard key metrics *FR-ADM-01*	1. Login as admin 2. Open Dashboard 3. Compare figures with the database/bookings list	Total bookings, active bookings, cancellations, occupancy rate, available rooms and revenue match the data		
TC-070	Charts by day, week and month *FR-ADM-02*	Switch the dashboard view between day, week and month	Charts/summaries update correctly for each period		
TC-071	Manage users - change role and deactivate *FR-ADM-03*	1. Open Users 2. Change a customer to staff 3. Deactivate another account	Role change and status saved; changes are logged		

TC-072	Deactivated user cannot log in (negative) *FR-ADM-03*	1. Deactivate a user 2. Try to login with that account	Login refused with an appropriate message		
TC-073	View all bookings and payments with filters *FR-ADM-04*	1. Open Bookings and Payments pages 2. Search and filter by status, date and user	Matching records displayed correctly		
TC-074	Export report *FR-ADM-05 (low)*	Click Export CSV and Export PDF on a report	Files download and contain the displayed data		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
K. Security, Usability and Compatibility					
TC-075	Input sanitisation (NoSQL injection / XSS) *NFR 4.2*	1. Enter {"$ne":null} style value in login fields via API 2. Enter <script>alert(1)</script> in name and guest fields	Injection attempt fails; script is not executed and is stored/shown as plain text		
TC-076	HTTPS and secrets *NFR 4.2*	1. Open the deployed site and inspect the URL and certificate 2. Check the repository for committed .env files	All traffic on HTTPS; JWT secret and DB credentials are not in the repository		
TC-077	Inline validation and error messages *NFR 4.4*	Submit forms with empty or invalid values	Clear inline messages and success/error notifications; loading indicators on slow actions		
TC-078	Protected route redirect *UI 5.1*	Without logging in, open My Bookings and Staff Console URLs	Redirected to Login page		
TC-079	Responsive layout *NFR 4.4*	1. Open the site on mobile and tablet widths 2. Walk through search, booking and payment	Pages adapt to screen size; all actions usable by touch		
TC-080	Browser compatibility *NFR 4.7*	Run the booking flow on latest Chrome, Firefox, Edge and Safari	Same behaviour and layout on all four browsers		

TC-081 Booking steps for first-time user
NFR 4.4

Ask a new user to find and book a room without help; note steps and time

User completes the booking unaided in a few steps

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001					
BG-002					
BG-003					
BG-004					

BG-005					
BG-006					

BG-007

BG-008

Sign-off:

Tester Name:

Date:

Signature:

Project Manager Sign-off:
Product Owner Sign-off:

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement. • Bug tracking should include details such as severity, status, and steps to reproduce. • Where the SRS does not define a business rule (cancellation and refund percentages, taxes, same-day check-out/check-in, hold time for pending bookings), record the observed behaviour and confirm the expected value with the team before marking Pass/Fail.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.